

RUSHALI MOTERIA

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Education

New York University

Master of Science in Computer Engineering

Relevant Coursework: Machine Learning, Deep Learning, Big Data

Sept 2024 – May 2026

GPA: 4.00/4.00

Ahmedabad University

Bachelor of Technology in Computer Science and Engineering

Relevant Coursework: Data Structures, Object-Oriented Programming, Software Engineering, Database Management

Aug 2020 – May 2024

GPA: 3.71/4.00

Experience

Software Engineering Intern (Infrastructure)

Sigma Computing

May 2025 – Present

NYC, USA

- Migrated Azure infrastructure from Terraform to Pulumi using Golang, modularizing over 8 resource types (DNS zone, AKS, MySQL, Redis) and **reducing provisioning errors by 30%**.
- Implemented stack-based Pulumi configurations for multi-subscription support and wrote Go-based unit tests with **80% coverage** to validate infrastructure logic and prevent misconfigurations.
- Automated CI/CD workflows using **GitHub Actions and ArgoCD** to enable seamless Pulumi preview, change detection, and deployment across 2 environments, ensuring consistent and auditable infrastructure updates.

Full Stack Developer Intern

Elluminati Inc.

April 2023 – July 2023

Rajkot, India

- Assisted in developing frontend components using React.js, ensuring functionality aligned with the design, and contributed to a project used by over **500+** active users.
- Worked with backend technologies like Node.js and Express.js to handle user authentication and API requests, improving response times for **endpoints by 20%**, measured using Postman benchmarks before and after code optimizations.
- Collaborated with the development team to test and debug features, ensuring smooth functionality for users.

Projects

Serverless Pipeline for Real-Time Inference on AWS | *Lambda, S3, API Gateway, DynamoDB, Docker* April 2025

- Built a fully event-driven serverless system that triggers model inference workflows on image uploads, ensuring stateless execution, automatic scaling, and minimal operational overhead.
- Optimized Lambda function performance by minimizing image preprocessing and reducing ONNX model load time, leading to faster inference execution and **lowering average API response latency by 30–40%**.
- Designed a fault-tolerant, asynchronous polling system via API Gateway and DynamoDB, achieving consistent **prediction retrieval within 3–5 seconds** across test uploads.

Efficient ResNet-18 for CIFAR-10 Classification | *PyTorch, NumPy, TensorFlow*

March 2025

- Designed a lightweight ResNet-18 model with under **5M parameters** for CIFAR-10 classification, optimizing accuracy while maintaining computational efficiency.
- Reduced parameter count by 50%** through architectural modifications like reducing convolutional filters and replacing standard residual blocks, while maintaining high classification performance.
- Achieved **93.7% validation accuracy** by applying **data augmentation techniques**, including random cropping, horizontal flipping, and color jitter, to improve model generalization.

Blockchain Escrow System | *Solidity, Next.js, GraphQL, Hardhat, Ethers.js, React.js*

May 2024

- Developed a secure and trustworthy blockchain-based escrow system on **Ethereum**, utilizing smart contracts to automate milestone-based payments, ensuring seamless and transparent transactions between parties.
- Utilized Hardhat for writing, testing, and deploying smart contracts, and optimized gas costs to **reduce transaction fees by 20%**.
- Integrated The Graph and GraphQL for decentralized data indexing and querying, enabling real-time data retrieval and updates, while performing unit testing with **80% code coverage** to ensure robust functionality and reliability.

Technical Skills

Programming Languages: C++, C, Python, JavaScript, SQL, Solidity, Go (Golang)

Cloud & Infrastructure: AWS (S3, Lambda, API Gateway), Microsoft Azure (AKS, Key Vault, DNS, MySQL, Redis), Terraform, Pulumi, Kubernetes

Web Development: ReactJS, Node.js, Express.js, MongoDB, PostgreSQL, HTML, CSS, AJAX, RESTful APIs, JWT

CI/CD: GitHub Actions, ArgoCD, Atlantis, Docker

ML/DL Frameworks: TensorFlow, PyTorch, Scikit-Learn, NumPy, Pandas

Database Management: MySQL, MongoDB, PostgreSQL, ER Diagrams, Data Modeling, Triggers, Stored Procedures